ROCKS & MINERALS TEST

Staten Island Technical High School
CSE 2025
17 stations (2 min 55 sec/station, 25 sec extra for last station)
Each question is 2 pts unless otherwise stated.

____/195

STATION 1 (11 pts)

A

B

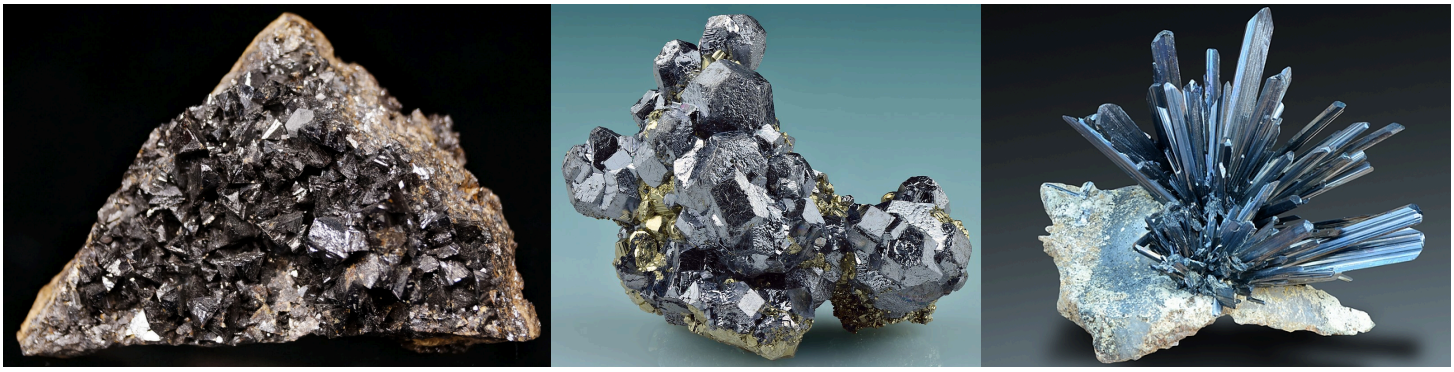
1. Is mineral A fluorescent? What color does it glow? Under what kind of radiation? (5)

2. Which mineral is the source of common salt?

3. Describe the diaphaneity of mineral B.

4. What kind of twinning does mineral A commonly exhibit?

STATION 2 (10 pts)



A B C

5. What is the index of refraction of mineral B?

6. What kind of fracture does mineral C have?

7. Describe one type of environment where mineral A might be found.

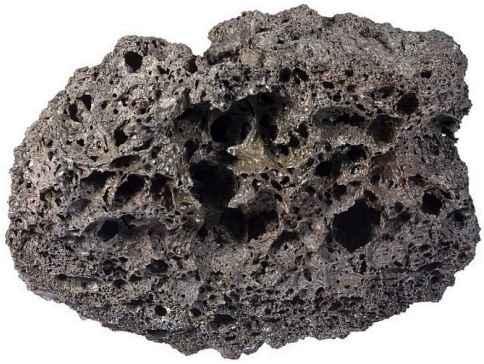
8. What was mineral A originally called when it was found in 1546?

9. Which two above minerals are piezoelectric?

STATION 3 (10 pts)

A

B



C

10. Describe the texture of rock A.

11. True or false: rock B is often magnetic.

12. Rock B is equivalent in composition to what other intrusive igneous rock?

13. Describe how the cavities in rock C typically form.

14. Rock A is mineral-like, but not a mineral. Explain why it is not classified as a mineral.

STATION 4 (10 pts)

A

B

15. True or false: rock A is the most common sedimentary rock. If false, correct the statement.

16. What rock will rock A metamorphose to in a low-temperature environment?

17. What is the streak of rock B?

18. Rock A is characterized by the ability to split into thin layers less than 1 cm in thickness. What is this property called?

19. What is the hardness of rock B?

STATION 5 (10 pts)

A

B

20. Identify these specimens.

21. Describe the fracture of specimen B.

22. True or false: specimen A has a low porosity. If false, correct the statement.

23. Who discovered specimen A? In what year? (4)

STATION 6 (10 pts)

A

B

24. What is the density of rock B (in g/cm³)?

25. Describe the luster of rock B.

26. List two uses of Rock A.

27. Rock A contains a certain mineral that can lead to cancer. Identify the mineral, and the type of cancer. (4)

STATION 7 (10 pts)

A

B

C

28. What is the chemical composition of mineral A?

29. Which mineral is Argentina's national gemstone?

30. What is the smell of mineral B?

31. What is the index of refraction of Mineral B?

32. Mineral A is anhydrous. What does it mean if a mineral is anhydrous?

STATION 8 (12 pts)

A

B

33. When mineral A is combusted, describe the visible product and the flame (4).

34. What distinctive characteristic does mineral A have to the human senses?

35. Another mineral (let's call it mineral C) on the *2025 Rocks and Minerals List* is often mistaken for mineral B. What is mineral C?

36. What use does mineral A have in gardening?

37. What unit of mass is utilized for measuring mineral B's purity?

STATION 9 (10 pts)

A

B

38. Describe the etymology of rock A, including the language, word of origin, and its meaning. (4)

39. Mineral B is the most common ore of which chemical element?

40. Explain why the pebbles in rock A are rounded.

41. Is rock B infusible?

STATION 10 (10 pts)

A

B

42. Mineral A has a quality where it becomes less soluble as temperature increases. What is this quality called?

43. Mineral B is often found with what other carbonate mineral?

44. What is the tenacity of mineral B?

45. Mineral A is etymologically related to which rock on the *2025 Rocks and Minerals List?*

46. What is the crystal system of mineral A?

STATION 11 (12 pts)

A

B

47. In what environment does rock A commonly form?

48. What is the hardness of rock B?

49. Ancient Egyptians made the statue of a mythological deity out of rock A. What is the name of this deity?

50. Does rock B often contain fossils?

51. Some specimens of rock A contain large rock fragments. What are these types of big rock formations called, and how do they form? (4)

STATION 12: BOWEN'S REACTION SERIES (16 pts)

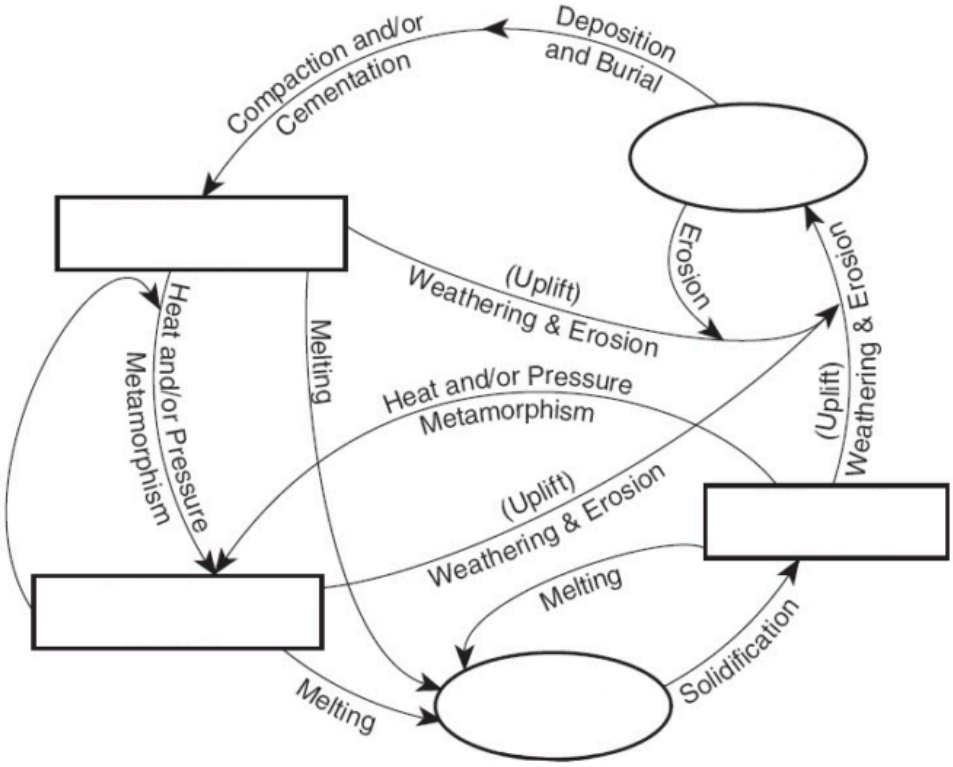
52. Draw a rough sketch of Bowen's reaction series (8).

53. What is the difference between the continuous (1) and discontinuous branches (1) in the series?

54. In what decades was Bowen's reaction series developed?

55. Why do certain minerals in the series crystalize at higher temperatures compared to others? (4)

STATION 13: THE ROCK CYCLE (13 pts)



56. Fill in the missing terms in the chart above. (5)

Describe each of the following processes in the rock cycle:

57. Uplift

58. Melting & solidification

59. Compaction & cementation

60. Erosion & deposition

STATION 14: IGNEOUS STRUCTURES (14 pts)

Match the igneous structure to its description, and determine if it is an intrusive or extrusive structure.

a. Batholith	b. Stock	c. Laccolith	d. Sill	e. Dyke	f. Volcanic Neck	g. Lava Plateau

61. ___ Tabular or sheetlike igneous body that is often oriented vertically or steeply inclined to the bedding of preexisting rocks. (Intrusive/Extrusive)

62. ___ Larger than 100 km² in area, that forms from cooled magma deep in the Earth's crust. (Intrusive/Extrusive)

63. ___ “Throat” of a volcano; consists of a pipelike conduit filled with hypabyssal rocks. (Intrusive/Extrusive)

64. ___ Has a surface exposure of less than 100 square kilometres; has a discordant relationship with the rocks that it intrudes. (Intrusive/Extrusive)

65. ___ Forms when magma solidifies between layers of existing rock, creating a tabular sheet parallel to the surrounding rock's bedding or foliation. (Intrusive/Extrusive)

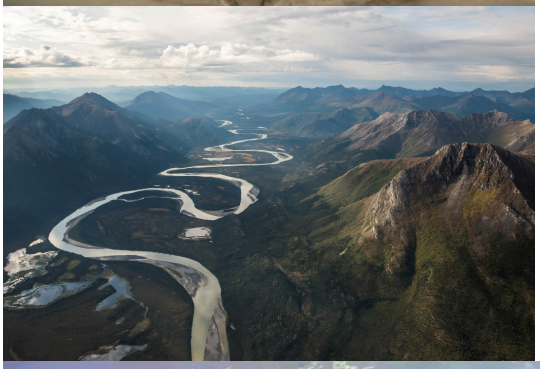
66. ___ Large, flat area formed by extensive, fluid lava flows that solidify into igneous rocks. (Intrusive/Extrusive)

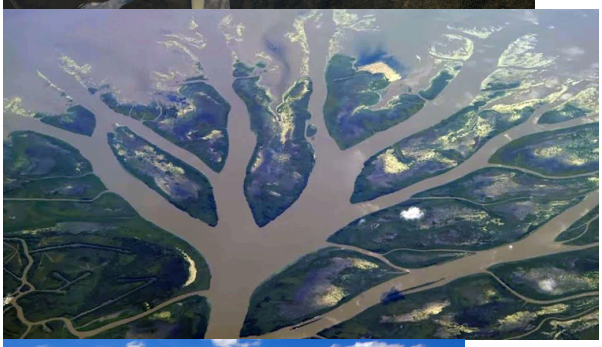
67. ___ Forms when magma forces its way into and between layers of sedimentary rock, lifting the overlying rock layers into a dome. (Intrusive/Extrusive)

STATION 15: ENVIRONMENTS OF DEPOSITION (12 pt)

Identify the environment of deposition from each photo.

68.

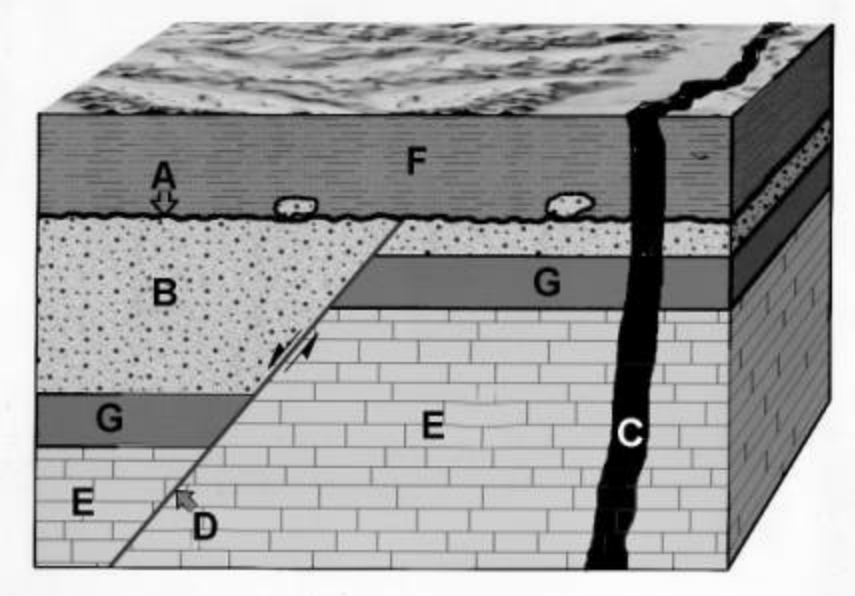
69. 

70. 

71.

72. Classify each landform above as continental, transitional, or marine (4).

STATION 16: RELATIVE DATING (15 pts)



73. Put the rock layers and unconformities in order, from oldest to youngest. (9)

74. Which law did you apply to determine if E or G was older/younger?

75. Which law did you apply to determine if C or F was older/younger?

76. Which law did you apply to determine if D or B was older/younger?

STATION 17: RAPID ID (10 pts; 1 pt each)

77. _____

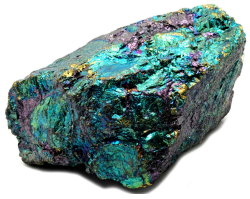
78. _____

79. _____

80. _____

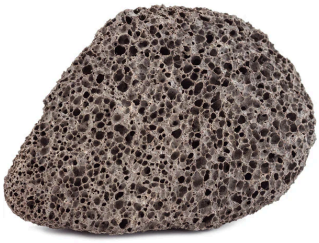
81. _____

82. _____



83. _____

84. _____



85. _____

86. _____